INTERSECTION NUMBERS ARE ADDITIVE IN EACH INVERTIBLE FACTOR

THE STACKS PROJECT AUTHORS

Editorial note. Counted span is the author's own proof environment, source lines 3404-3416, immediately following the statement at 3390-3402.

Editorial note. The statement's displayed additivity formula is part of the counted statement span; the polynomiality input is cited at proof lines 3404-3406.

Editorial note. One same-chapter lemma stub and one definitional stub are recorded in assembly.external_references.

Allowed standard prerequisites (editorial).

- Intersection number as an Euler characteristic polynomial. Definition definition-intersection-number of this chapter, cited at statement line 3391. Fixes the definition: the intersection number is the coefficient extraction from the Euler characteristic power series.
- Numerical polynomiality of the Euler characteristic in tensor powers. Lemma lemma-numerical-polynomial-from-euler of this chapter, cited at proof lines 3404-3406. The function of d+1 variables (n_1..n_{i-1}, n_i', n_i", n_{i+1}..n_d) built from chi(Z, tensor of powers restricted to Z) is a numerical polynomial of total degree at most d.

Source reference. Original definition 0EDF: sources/stacks-spaces-over-fields/pinned-original-source.tex, source lines 3348–3348. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 0DN4: sources/stacks-spaces-over-fields/pinned-original-source.tex, source lines 3053–3053. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *In the situation of Definition 0EDF the intersection number* $(\mathcal{L}_1 \cdots \mathcal{L}_d \cdot Z)$ *is additive: if* $\mathcal{L}_i = \mathcal{L}_i' \otimes \mathcal{L}_i''$*, then we have*

$$(\mathcal{L}_1 \cdots \mathcal{L}_i \cdots \mathcal{L}_d \cdot Z) = (\mathcal{L}_1 \cdots \mathcal{L}_i' \cdots \mathcal{L}_d \cdot Z) + (\mathcal{L}_1 \cdots \mathcal{L}_i'' \cdots \mathcal{L}_d \cdot Z)$$

Proof. This is true because by Lemma 0DN4 the function

$$(n_1, \ldots, n_{i-1}, n_i', n_i'', n_{i+1}, \ldots, n_d) \mapsto \chi(Z, \mathcal{L}_1^{\otimes n_1} \otimes \ldots \otimes (\mathcal{L}_i')^{\otimes n_i'} \otimes (\mathcal{L}_i'')^{\otimes n_i''} \otimes \ldots \otimes \mathcal{L}_d^{\otimes n_d}|_Z)$$

is a numerical polynomial of total degree at most d in $d+1$ variables. □

Date: stacks-project:spaces-over-fields.tex:local-snapshot-20261012; GNU FDL 1.2 or later.